

TR-APP-001

- Related Functional Requirement: FR-APP-001
- Objective: Implement data ingestion from CSV source and load into Delta tables for customer and order.
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
 - customer: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
 - order: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Target Data Details:
 - Delta tables: customer, order
- Job Flow:
 1. Trigger job to read CSV files from specified paths
 2. Load customer CSV into customer Delta table
 3. Load order CSV into order Delta table
- Transformations & Logic:
- Error Handling & Logging:
 - Log missing file or read errors
 - Log successful completion or failures

TR-APP-002

- Related Functional Requirement: FR-APP-002
- Objective: Enforce correct schema for customer and order Delta tables.

- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- Raw CSV files
- Target Data Details:
- customer: CustId, Name, EmailId, Region
- order: OrderId, ItemName, PricePerUnit, Qty, Date, CustId
- Job Flow:

1. Load data as per defined schema

2. Enforce schema on table creation

- Transformations & Logic:
- Error Handling & Logging:
- Schema enforcement errors logged

TR-APP-003

- Related Functional Requirement: FR-APP-003
- Objective: Add 'TotalAmount' column to order table as $\text{PricePerUnit} * \text{Qty}$.
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:

- Auto Termination:
 - Libraries:
 - Source Data Details:
 - order table
 - Target Data Details:
 - order table with TotalAmount column
 - Job Flow:
1. Read loaded order data
 2. Calculate TotalAmount for each record
 3. Update table with new column
- Transformations & Logic:
 - $\text{TotalAmount} = \text{PricePerUnit} * \text{Qty}$
 - Error Handling & Logging:
 - Log calculation errors or data inconsistencies

TR-APP-004

- Related Functional Requirement: FR-APP-004
- Objective: Remove null and duplicate records from both customer and order tables.
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- customer and order delta tables
- Target Data Details:

- Cleaned customer and order tables
- Job Flow:
 1. Scan for null records
 2. Remove records with null values
 3. Detect and eliminate duplicate records
- Transformations & Logic:
 - Null value exclusion
 - Duplicate removal
- Error Handling & Logging:
 - Log number of records removed and reason

TR-APP-005

- Related Functional Requirement: FR-APP-005
- Objective: Create ordersummary table in specified catalog/schema if not exists, with correct schema.
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
- Target Data Details:
 - ordersummary table with expected columns
- Job Flow:
 1. Check for table existence
 2. Create table if missing, with specified schema

- Transformations & Logic:
- Error Handling & Logging:
- Log table creation success or errors

TR-APP-006

- Related Functional Requirement: FR-APP-006
- Objective: Join customer and order tables on CustId and populate ordersummary as SCD type 2.
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
 - customer table, order table
- Target Data Details:
 - ordersummary table
- Job Flow:

1. Perform inner join on CustId
2. Load joined data to ordersummary, manage SCD2 logic

- Transformations & Logic:
- SCD2 mechanics for inserts/updates
- Error Handling & Logging:
- Log number of loaded and updated records

TR-APP-007

- Related Functional Requirement: FR-APP-007
- Objective: Update ordersummary SCD type 2 with logic to handle inactivity, new activity, and date maintenance.
- Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries:
- Source Data Details:
 - ordersummary, customer tables
- Target Data Details:
 - Updated ordersummary table
- Job Flow:
 1. Monitor customer table for changes
 2. On change, set old records to Inactive in ordersummary
 3. Insert new record as Active, update StartDate and EndDate
- Transformations & Logic:
 - SCD type 2 update operations
- Error Handling & Logging:
 - Log all SCD2 update events

TR-APP-008

- Related Functional Requirement: FR-APP-008
- Objective: Create customeraggregatespend table in specified catalog/schema if not exists.
- Target Cluster Configuration:
 - Cluster Name:

- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:
- Target Data Details:
- customeraggregatespend(Name, TotalAmount, Date)
- Job Flow:

1. Check for table existence

2. Create if not found

- Transformations & Logic:
- Error Handling & Logging:
- Log table creation success/errors

TR-APP-009

- Related Functional Requirement: FR-APP-009
- Objective: Aggregate TotalAmount per Name and Date from ordersummary and load results into customeraggregatespend.
- Target Cluster Configuration:
- Cluster Name:
- Runtime Version:
- Node Type:
- Driver Node:
- Worker Nodes:
- Autoscaling:
- Auto Termination:
- Libraries:
- Source Data Details:

- ordersummary table
- Target Data Details:
- customeraggregatespend table
- Job Flow:

1. Aggregate by Name and Date, sum TotalAmount

2. Load aggregation into customeraggregatespend

- Transformations & Logic:
- groupBy aggregation
- Error Handling & Logging:
- Log aggregates and loading actions

TR-APP-010

- Related Functional Requirement: FR-APP-010
 - Objective: Implement entire end-to-end workflow using Delta Live Tables.
 - Target Cluster Configuration:
 - Cluster Name:
 - Runtime Version:
 - Node Type:
 - Driver Node:
 - Worker Nodes:
 - Autoscaling:
 - Auto Termination:
 - Libraries: Delta Live Tables
 - Source Data Details:
 - All tables in workflow
 - Target Data Details:
 - All target tables in workflow
 - Job Flow:
1. Develop pipeline using Delta Live Tables for steps TR-APP-001 to TR-APP-009

- Transformations & Logic:
- Use of Delta Live Tables orchestrator
- Error Handling & Logging:
- Log pipeline execution steps, failures, and successes